Ian Kronquist

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C, C++, Rust, Python, iOS/MacOS, Windows, Linux, Hypervisors, Kernels

Employment

Apple, Kernel Systems Red Team & Coprocessors Red Team *Senior Security Researcher November 2021-Present*

Found vulnerabilities in the L4 microkernel which powers Apple's Secure Enclave Processor. Reviewed and hardened coprocesser firmware and kernel drivers. Cooperated closely with hardware and software teams to design and review a system to securely paint content on the display even in the presence of a compromised XNU kernel.

Microsoft, Platform Sec. & Kernel Sec. Teams *Software Engineer II September 2017-August 2021*

Designed and implemented mitigations to harden Windows components and break exploit techniques. One such mitigation, Win32k Object Type Isolation, successfully defended against over a dozen CVE-worthy cases in 2019 alone. Developed CastGuard, a compiler based mitigation to prevent illegal downcasts in C++. Helped deliver Address Sanitizer support for the Microsoft MSVC C++ compiler. Became the maintainer of the Windows heap used throughout the OS and improved heap performance, security, and debugging information.

Barkly Protects, Protection Team *Software Engineer July 2016-July 2017*

Core developer of a hypervisor and Windows filesystem filter driver designed to detect and stop malware. Collaborated with malware researchers to develop new rules to stop ransomware and measure protection coverage using the Cuckoo malware analysis framework.

Education

Oregon State University *Corvallis, Oregon September 2013-June 2017*

Bachelor of Science, Computer Science, Honors Degree. Implemented a honeypot to monitor vulnerable containers as a Senior Project.

AFS Turkey Foreign Exchange Program *Adana, Turkey September 2012-July 2013*

Attended Gündoğdu Koleji (Sunrise High School) in Adana, Turkey. Proficient in the Turkish Language.

Eagle Scout

Designed, constructed, and gathered donations for a lending library as an Eagle Project.

Talks

The Life & Death of Kernel Object Abuse by Type Isolation *OPCDE Dubai & Hack In the Box Amsterdam*